

Algebra II with Trigonometry

Chapter 7.3

Olympiad Solutions

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[4592201_alg-2trig-h-chp-7-3-note-docx.pdf](#)

Solutions

1. Given that $\sin \theta + \sin 3\theta + \sin 5\theta = 0$, find the exact value of $\cos 2\theta \cdot \cos 4\theta$.

Solution:

We use the sum-to-product identities:

$$\sin \theta + \sin 5\theta = 2 \sin \left(\frac{\theta + 5\theta}{2} \right) \cos \left(\frac{\theta - 5\theta}{2} \right) = 2 \sin 3\theta \cos(-2\theta) = 2 \sin 3\theta \cos 2\theta$$

Our equation becomes:

$$\sin \theta + \sin 3\theta + \sin 5\theta = (\sin \theta + \sin 5\theta) + \sin 3\theta = 2 \sin 3\theta \cos 2\theta + \sin 3\theta = \sin 3\theta (2 \cos 2\theta + 1) = 0$$

So either:

- $\sin 3\theta = 0$, or
- $2 \cos 2\theta + 1 = 0 \implies \cos 2\theta = -\frac{1}{2}$

Let's examine the second factor, as for $\sin 3\theta = 0$ the value of $\cos 2\theta \cdot \cos 4\theta$ will depend on θ .

If $\cos 2\theta = -\frac{1}{2}$:

- $\cos 2\theta = -\frac{1}{2}$
- $2\theta = 120^\circ$ or 240° , so $\theta = 60^\circ$ or 120°

Now compute $\cos 4\theta$:

- For $\theta = 60^\circ$: $4\theta = 240^\circ$, $\cos 240^\circ = -\frac{1}{2}$
- For $\theta = 120^\circ$: $4\theta = 480^\circ = 480 - 360 = 120^\circ$, $\cos 120^\circ = -\frac{1}{2}$

Thus, for each case,

$$\cos 2\theta \cdot \cos 4\theta = \left(-\frac{1}{2}\right) \left(-\frac{1}{2}\right) = \boxed{\frac{1}{4}}$$

2. If $\tan x = \sqrt{7} + 2\sqrt{3}$, compute $\sin 2x$ in simplest radical form.

Solution:

Let $t = \tan x$:

$$\sin 2x = \frac{2 \tan x}{1 + \tan^2 x} = \frac{2t}{1 + t^2}$$

Plug in $t = \sqrt{7} + 2\sqrt{3}$:

First, compute t^2 :

$$t^2 = (\sqrt{7} + 2\sqrt{3})^2 = 7 + 4\sqrt{21} + 12 = 19 + 4\sqrt{21}$$

So:

$$1 + t^2 = 1 + 19 + 4\sqrt{21} = 20 + 4\sqrt{21}$$

and

$$2t = 2(\sqrt{7} + 2\sqrt{3}) = 2\sqrt{7} + 4\sqrt{3}$$

Put together:

$$\sin 2x = \frac{2\sqrt{7} + 4\sqrt{3}}{20 + 4\sqrt{21}} = \frac{2\sqrt{7} + 4\sqrt{3}}{4(5 + \sqrt{21})} = \frac{\sqrt{7} + 2\sqrt{3}}{2(5 + \sqrt{21})}$$

Rationalize the denominator:

$$\frac{\sqrt{7} + 2\sqrt{3}}{2(5 + \sqrt{21})} \cdot \frac{5 - \sqrt{21}}{5 - \sqrt{21}} = \frac{(\sqrt{7} + 2\sqrt{3})(5 - \sqrt{21})}{2((5)^2 - (\sqrt{21})^2)}$$

Compute numerator:

$$(\sqrt{7} + 2\sqrt{3})(5 - \sqrt{21}) = 5\sqrt{7} - \sqrt{7}\sqrt{21} + 10\sqrt{3} - 2\sqrt{3}\sqrt{21} = 5\sqrt{7} - \sqrt{147}$$

$$\text{But } \sqrt{147} = 7\sqrt{3}, \text{ and } 2\sqrt{3}\sqrt{21} = 2\sqrt{63} = 6\sqrt{7}:$$

So numerator:

$$5\sqrt{7} - 7\sqrt{3} + 10\sqrt{3} - 6\sqrt{7} = (5\sqrt{7} - 6\sqrt{7}) + (10\sqrt{3} - 7\sqrt{3}) = -\sqrt{7} + 3\sqrt{3}$$

Denominator:

$$2(25 - 21) = 2 \cdot 4 = 8$$

Thus,

$$\boxed{\sin 2x = \frac{3\sqrt{3} - \sqrt{7}}{8}}$$

3. Without using a calculator, evaluate $\frac{\sin 75^\circ + \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}$.

Solution:

Use the sum-to-product formulas:

$$\sin A + \sin B = 2 \sin \left(\frac{A + B}{2} \right) \cos \left(\frac{A - B}{2} \right)$$

$$\cos A + \cos B = 2 \cos \left(\frac{A + B}{2} \right) \cos \left(\frac{A - B}{2} \right)$$

So,

$$\sin 75^\circ + \sin 15^\circ = 2 \sin(45^\circ) \cos(30^\circ)$$

$$\cos 75^\circ + \cos 15^\circ = 2 \cos(45^\circ) \cos(30^\circ)$$

Thus,

$$\frac{\sin 75^\circ + \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ} = \frac{2 \sin 45^\circ \cos 30^\circ}{2 \cos 45^\circ \cos 30^\circ} = \frac{\sin 45^\circ}{\cos 45^\circ} = \tan 45^\circ = \boxed{1}$$

4. Suppose $\sin^4 x + \cos^4 x = \frac{5}{8}$. Find the value of $\sin^2 2x$.

Solution:

$$\text{Let } S = \sin^4 x + \cos^4 x.$$

We know:

$$S = (\sin^2 x)^2 + (\cos^2 x)^2 = (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x = 1 - 2 \sin^2 x \cos^2 x$$

Given $S = \frac{5}{8}$, so:

$$1 - 2 \sin^2 x \cos^2 x = \frac{5}{8} \implies 2 \sin^2 x \cos^2 x = 1 - \frac{5}{8} = \frac{3}{8}$$

But:

$$\sin^2 2x = 4 \sin^2 x \cos^2 x = 2 \times 2 \sin^2 x \cos^2 x = 2 \times \frac{3}{8} = \frac{3}{4}$$

So the answer is $\boxed{\frac{3}{4}}$.

5. Let $t = \tan^{-1} \frac{1}{3}$. Find the exact value of $\sin 2t + \cos 2t$.

Solution:

$$\text{Let } t = \tan^{-1} \frac{1}{3}.$$

Draw a right triangle: opposite = 1, adjacent = 3, hypotenuse
 $= \sqrt{1^2 + 3^2} = \sqrt{10}.$

So:

$$\sin t = \frac{1}{\sqrt{10}}, \quad \cos t = \frac{3}{\sqrt{10}}$$

Compute $\sin 2t$ and $\cos 2t$:

$$\sin 2t = 2 \sin t \cos t = 2 \times \frac{1}{\sqrt{10}} \times \frac{3}{\sqrt{10}} = \frac{6}{10} = \frac{3}{5}$$

$$\cos 2t = 2 \cos^2 t - 1 = 2 \left(\frac{3}{\sqrt{10}} \right)^2 - 1 = 2 \times \frac{9}{10} - 1 = \frac{18}{10} - 1 = \frac{8}{10} = \frac{4}{5}$$

Sum:

$$\sin 2t + \cos 2t = \frac{3}{5} + \frac{4}{5} = \boxed{\frac{7}{5}}$$

6. If $0 < x < \frac{\pi}{2}$ and $\cos x = 1 - 2 \sin^2 \frac{x}{2}$, determine the value of $\sin x$.

Solution:

Recall the double angle identity:

$$\cos x = 1 - 2 \sin^2 \frac{x}{2}$$

But also, using the double angle formula:

$$\cos x = 1 - 2 \sin^2 \frac{x}{2}$$

which is identical to the given equation.

Thus, the equation is always valid for all x in $0 < x < \frac{\pi}{2}$. Let's compute $\sin x$ in terms of $\sin \frac{x}{2}$:

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

But we can also write, using Pythagoras:

$$\left(\sin \frac{x}{2}\right)^2 + \left(\cos \frac{x}{2}\right)^2 = 1$$

Let $y = \sin \frac{x}{2}$, then $\cos \frac{x}{2} = \sqrt{1 - y^2}$ (since $0 < x < \pi/2 \implies 0 < \frac{x}{2} < \frac{\pi}{4}$, both positive).

So

$$\sin x = 2y\sqrt{1 - y^2}$$

But all we are asked is: what is $\sin x$? Is there any constraint? No. The identity holds for all x in $0 < x < \frac{\pi}{2}$, so there is **no further restriction** and any value of $\sin x$ in $(0, 1)$ is possible.

But sometimes this question intends for you to realize:

Let's check from the double-angle:

$$\cos x = 1 - 2 \sin^2 \frac{x}{2} = \cos x$$

So indeed, $\sin x$ can be **any** value in $(0, 1)$.

However, if we double-check: perhaps there is an alternative.

But let's solve:

Set $y = \sin \frac{x}{2}$.

Given $\cos x = 1 - 2 \sin^2 \frac{x}{2}$.

But $\cos x = 1 - 2 \sin^2 \frac{x}{2}$, so the given equation is just an identity.

Thus, any $\sin x$ between $(0, 1)$ is valid.

Perhaps the problem meant to check for a value—we try x that fits a "nice" sine value. Let's try $x = 0$ (But $x > 0$), so not allowed.

Alternatively, perhaps the answer is just:

$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$
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If more is intended, perhaps we're to use a special value. But with only this equation, the general solution is as above. If the intention is to recognize the double angle:

$$\sin^2 \frac{x}{2} = y^2$$

$$\sin x = 2y\sqrt{1 - y^2}$$

Thus, the answer is $\boxed{\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}}$.

Alternatively, since the identity holds for all x in the interval, there is no fixed value for $\sin x$. (If any further specification, please clarify.)
